

Tilt switch

DETECT

INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

A tiny metal ball rolls inside and touches two contacts, completing the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses



Box lid sensor

DETECT
Lid changes
angle

DECIDE
Box has
opened

DO
Log or sound
an alert



Grumpy pencil case

DETECT
Case is tipped

DECIDE
Pencils are
escaping

DO
Make a
grumpy noise



Metal can
Contains the
rolling ball

Legs
No polarity

Angle
Changes the
contact

Code

● Raspberry Pi

```
from gpiozero import Button

tilt = Button(23)

tilted = tilt.is_pressed
```

● Raspberry Pi Pico

```
from picozero import Switch

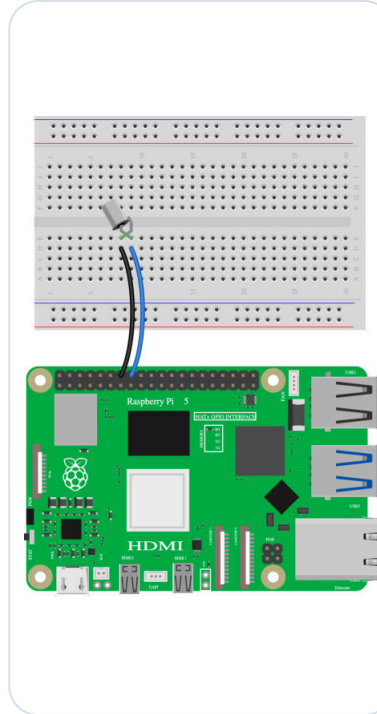
tilt = Switch(16)

tilted = tilt.is_closed
```

The reading may flicker while the ball rolls.
Check it more than once before treating it as a definite tilt.

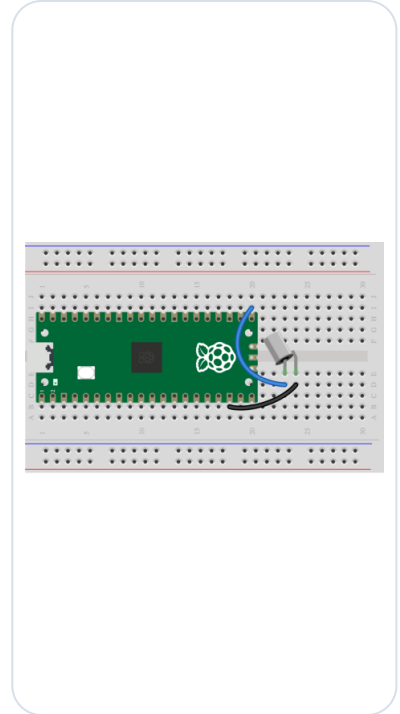
Wiring

RASPBERRY PI 5



GPIO 23 → tilt switch → GND

PICO



GP16 → tilt switch → GND